

- 16** The table compares the surface area to volume ratios of five agar blocks that differ in dimensions but which all have the same volume.

The agar blocks can be used to measure the efficiency of diffusion, where efficiency is measured as the time taken for a dye to reach all parts of the block.

	length / mm	width / mm	height / mm	surface area / mm ²	volume / mm ³	surface area : volume ratio
1	8	8	8	384	512	0.75
2	16	16	2	640	512	1.3
3	32	4	4	544	512	1.1
4	32	32	0.5	2112	512	4.1
5	64	4	2	784	512	1.5

Which prediction can be made about the way in which size and dimensions of these blocks affect the efficiency of diffusion?

- A** The efficiency of diffusion will decrease as the width of a block increases.
- B** The efficiency of diffusion will increase as the height of a block increases.
- C** The efficiency of diffusion will increase as a block of fixed volume is flattened.
- D** The efficiency of diffusion will decrease as a block of fixed volume is elongated.